

Summary of "Rearchitecting Datacenter Lifecycle for AI: A TCO-Driven Framework"

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Problem and Motivation

The paper addresses the high cost and inefficiency of managing AI datacenters for large language model (LLM) inference, where traditional datacenter lifecycle practices struggle to keep up with rapidly evolving models, growing resource demands, and expensive GPU hardware. The motivation for addressing this problem comes from the explosive growth of LLM usage across industries, which drives unprecedented infrastructure demand and creates a pressing need to manage costs, power, and resources effectively while supporting increasingly complex AI workloads.

Related Works

Some related work on cross-stage optimizations in AI datacenters includes:

- **IT provisioning → Build:** NVIDIA introduced liquid cooling to handle higher-density AI workloads and adopted InfiniBand and NVLink networking to support higher bandwidth and lower latency. These changes increase initial build costs but help extend the usable lifetime of datacenter deployments by easing future hardware refreshes.
 - **Operate → IT provisioning:** Research such as Helix and Splitwise investigates heterogeneity-aware scheduling, allowing older GPUs to be assigned to memory- or bandwidth-intensive tasks while reserving newer GPUs for compute-heavy phases, which helps reduce refresh costs and sustain high hardware utilization.
 - **Build → Operate:** Prior work shows that infrastructure decisions made during build time can enable software controls to balance demand and sustain efficiency. POLCA demonstrates that LLM inference clusters can safely oversubscribe power to provision up to 30% more servers with minimal performance loss, while TAPAS adds thermal and power-aware scheduling to dynamically manage workloads and handle emergencies.
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Solution Overview

A datacenter's lifecycle consists of three main stages: building, IT provisioning, and operation. To assess the long-term economics of AI datacenters, the authors develop a TCO-driven framework that evaluates costs over a 15-year lifecycle. This framework captures how design, deployment, and operational decisions influence total cost over time. It integrates factors such

as workload growth, model evolution, hardware advancements, and infrastructure expenses to simulate alternative lifecycle policies. By modeling both Capital Expenditures (CapEx) including facility construction, networking, and hardware, and Operational Expenditures (OpEx) such as energy, maintenance, and software, the framework provides a comprehensive view of how different strategies affect the overall Total Cost of Ownership (TCO) for AI datacenters.

Category	Component	Description	Example Cost (\$)
CapEx	IT	Servers, racks, accelerators, storage	\$375k/server [18]
	Networking	Fabric switches, optics, structured cabling	\$2000/server [59, 85]
	Building	Site preparation, building shell, land, electrical and mechanical base infrastructure	\$0.5/ft ² [29]
	Power	Power infrastructure (Switchgear, transformers, UPS, PDUs, busbars, rack distribution)	\$7.0/W [26, 116]
	Cooling	Cooling infrastructure (chillers, CRAH/CRAC units, pumps, piping, liquid loops, airflow)	\$2.5/W [103, 116]
OpEx	Networking	Port licenses, optics replacement, networking component power	\$600/server [59, 82, 84]
	Energy	IT load scaled by PUE, utility tariffs, demand charges	\$20–40/MWh [35]
	Maintenance	Spares, repairs, monitoring, water/treatment, field-replaceable units, failure-rate	\$5000/server [102]
	Software	Licenses, support contracts	\$200/server

Table 3. TCO components for an example datacenter with DGX H100 [18] servers.

Build Stage: AI workloads are reshaping datacenter design due to their extreme power densities, thermal requirements, and interconnect needs. Using a TCO-driven framework, the authors analyze how to redesign infrastructure for efficiency across power, cooling, and networking systems.

- **Power Infrastructure:** Traditional hierarchical power delivery creates stranded capacity, leading to inefficiency. Flatter power distribution architectures can reduce fragmentation with trade-offs in fault tolerance, design and maintenance complexity. TCO-driven analysis shows that adopting flatter power distribution architectures improves overall power utilization and lowers lifecycle TCO by 4.2%.
- **Cooling Infrastructure:** Air cooling becomes inadequate for dense GPU racks, which can generate up to 8× more heat than CPU racks. Hybrid cooling systems combining air and liquid approaches achieve the best trade-off, lowering lifecycle TCO by 9%.
- **Networking Infrastructure:** AI models require high interconnect bandwidth and low latency. A hierarchical networking design (NVLink within a server, InfiniBand within a rack, and Ethernet across racks) cuts TCO by 6%.

IT Provisioning Stage: After datacenter construction, managing the AI hardware lifecycle is critical to maintain performance, efficiency, and scalability. Using a TCO-driven framework, the authors evaluate refresh strategies, hardware trends, and workload evolution to optimize GPU deployment.

- **GPU Trends:** Modern GPUs deliver vastly higher compute throughput and memory bandwidth compared to older generations, but come at higher costs. Performance, power efficiency, and cost-effectiveness vary by model size and architecture, with sparse and state-space models extending the utility of legacy hardware.
- **Refresh Policies:** Traditional fixed refresh intervals are suboptimal for AI workloads. Flexible strategies like retiring older GPUs when efficiency gains are high, extending life when gains are marginal, or skipping intermediate generations can reduce TCO by 15–20%.

- **Workload Evolution:** Rapidly growing AI models require accelerated refresh cycles to meet compute demands, whereas stable or smaller models allow extended hardware lifetimes.

Operation Stage: Once hardware is deployed, sustaining high utilization while meeting SLOs requires software strategies that adapt to evolving workloads and heterogeneous hardware. TCO-driven analysis shows that operations play a critical role in overall efficiency.

- **Traditional Approach:** General-purpose datacenters run workloads on homogeneous hardware, migrating to the latest servers to maximize throughput and power efficiency. Uniform operational strategies suffice due to predictable workloads and linear performance scaling.
 - **Rearchitecting for AI:** AI workloads and GPUs scale non-linearly, making direct hardware migration suboptimal. Software techniques like model migration smoothing, quantization, KV-cache management, and disaggregation optimize compute, memory, and hardware utilization.
 - **Heterogeneity-Aware Scheduling:** Routing workloads to the best-suited hardware generation and aligning operations with datacenter constraints extends hardware life, defers refreshes, and improves efficiency.
 - **TCO Impact:** Individual optimizations reduce TCO by 12–39%, while combining all strategies achieves over 60% reduction.
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Limitations

Some limitations of the research include its reliance on assumed model growth trends and benchmarks rather than real-world deployments and a focus on specific GPU vendors that may limit generalizability. Additionally, implementing proposed cross-stage optimizations may incur significant engineering overhead, and external factors like power pricing or supply constraints are not fully considered.

Future Research Directions

The authors mention future research directions in Cross-Stage Optimizations, emphasizing the co-design of infrastructure, hardware, and software with lifecycle-aware strategies. On the infrastructure side, they suggest exploring mixed-generation racks and heterogeneous derating to better leverage diverse hardware and reduce interconnect and power inefficiencies. For hardware, future work could focus on designing accelerators that balance high performance with long-term compatibility, lower power density, and specialized units like Processing-in-Memory (PIM) to handle distinct computational phases efficiently. For operation, research could investigate dynamic techniques such as KV-cache management, memory-aware placement, and novel model architectures (e.g., MoEs) to align workload demands with heterogeneous fleets.

Summary of Class Discussion

Q: How might the proposed TCO-driven framework adapt if future AI workloads shift from GPU-dominated architectures to specialized accelerators like TPUs or custom ASICs?

A: Would have to redo the analysis for the new accelerators.

Q: Is the cooling configuration (75% liquid, 25% air) within a rack? Does this imply that the workload would have to be distributed such that some work would be done in air cooled servers and more intense work would be done in liquid cooled?

A: There will be liquid cooled plates distributed between the racks and there would also be air cooling that would be circulating that cold air up through the racks.

Q: In the plot (figure 13) why does the number of servers go down, shouldn't it be similar? Or are they expecting that the demand would go down?

Rearchitecting Datacenter Lifecycle for AI:
A TCO-Driven Framework

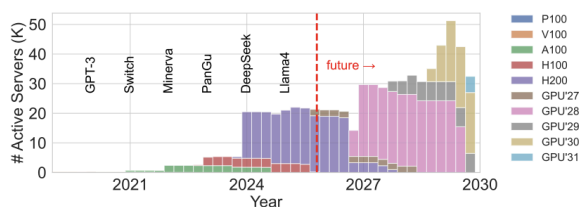
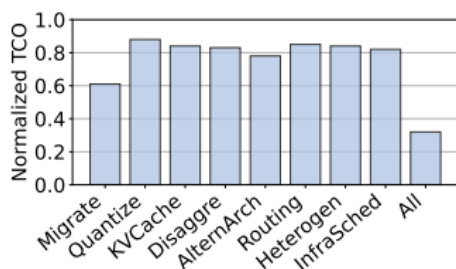


Figure 13. Server count by GPU type over time in an AI fleet following the optimal refresh policy for minimizing TCO.

A: They are expecting that the gpu's will be better so we will need the same amount of servers.



(a) TCO vs. software optimization.

Q: How did they calculate these numbers (figure 15a) because there are so many techniques? Especially with something like migration

A: I think it's using their simulator. They don't go into details. Best guess is you know the number of weights of the old model and new model and calculate with that.

Q: What is the input in the simulator?

A: Input would be the roofline model and account for changes

Q: But they're claiming that this tco/software improvement is much better than other improvements (cooling, etc.)?

A: Not sure. It's easier to show software gains.

Q: When it was talking about the two different operations, is there a reason they chose 15-30 years for the life cycle?

A: They're assuming that by 2030 the major data centers are going to have been built in 2015. Based on historical experience: Ricardo Bianchini runs data centers for Microsoft. They have an idea of how long these sites will run.

Summary of “Insights into DeepSeek-V3: Scaling Challenges and Reflections on Hardware for AI Architectures”

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Problem and Motivation

This paper addresses how hardware aware model co-design can alleviate a lot of current hardware architectures limitations such as memory capacity, computational efficiency, and interconnection bandwidth. Making models larger has shown great improvement in model performance. However, model sizes have reached a level where massive infrastructure and tens to hundreds of thousands of GPUs are needed to train and run inference. This poses two challenges, hardware improvements cannot keep up with this continuous increase in compute demand and smaller teams cannot compete with larger companies that have access to this infrastructure. To address these issues, the authors discuss techniques used in creating the DeepSeek-V3 model to maximize limited hardware, ranging from innovations in model architecture with the mixture of experts and multi-headed latent attention approach as well improvements in network topology to minimize network overhead.

Related Works

Prior research on scaling LLMs focused on improving memory efficiency, computational throughput, and interconnect performance. Techniques such as Grouped-Query Attention and Multi-Query Attention have aimed to reduce KV-cache memory, while windowed attention mechanisms and KV quantization further address memory-bound decoding bottlenecks. Mixture-of-Experts architectures, popularized by works like GShard and Switch Transformer, have shown that sparse activation enables large parameter models with reduced compute. Efforts in low-precision computation such as FP8 quantization, GPTQ, and AWQ have explored reducing memory and computation during inference, though no other large models used FP8 for training.

Solution Overview

The key innovation in DeepSeek-V3 is its hardware-aware design. This begins with its usage of a mixture of expert architecture where only a fraction of the total parameters are activated per token allowing a larger model to run with less available compute. As models grow in size and memory demand grows, GPU shared memory has not been able to keep up. To address this, DeepSeek-V3 also leverages multi-head latent attention to compress KV cache entries as well as lower precision data types to significantly reduce the memory pressure. The

authors also speed up inference using dual microbatch overlapping, a technique where computation is overlapped with communication ensuring the GPU remains fully utilized. Multi-token prediction is also used where at every step additional candidate tokens are generated at lower cost and verified in parallel allowing inference end to end generation latency to drop at a small cost of throughput.

In order to address the increasing memory pressure created by larger models, the authors explore various floating point data types. An innovation that the authors made was using FP8 mixed precision during training, which had not been done in big models previously. Some hardware limitations that the authors mention are the limited accumulation precision affecting training stability and the nontrivial overhead to quantize and dequantize incurring frequent data movements and reducing computational efficiency. The authors also experimented with a logarithmic floating point format that mapped activations to a log space instead of a linear space. However, this overhead proved to be too large with the log and exponent operations.

Another challenge that the authors faced was limited interconnect bandwidth due to regulatory compliance. This meant that they had less than half of a regular H800's NVLink bandwidth. Therefore, DeepSeek-V3's parallelism strategy accounted for this limitation by avoiding TP during training, enhancing pipeline parallelism with DualPipe, and accelerating expert parallelism by relying on InfiniBand NICs for inter-node communication. Because of the discrepancy between intra-node and inter-node bandwidth, the authors employ a Node-Limited routing where experts in the mixture of experts strategy is placed into groups and each group is placed in one node. This ensures that IB transfer only happens once and NVLink transfer can be used thereafter. An area of hardware improvement that the authors mention is transferring communication related operations to dedicated hardware components like InfiniBand does with NICs. This would unify the two frameworks. Another hardware limitation that the authors mention is dynamic bandwidth allocation between different types of traffic. Currently, KV cache transfer from CPU memory to GPU memory can conflict with IB communication and this can degrade the performance of both. The authors also propose CPUs and GPUs to be connected using NVLink instead of just PCIe.

My favorite part of the paper was the co-design with network topology in mind. The authors built a multi plane fat tree topology where each GPU, NIC pair was assigned to a different plane. This topology had benefits of being able to take advantage of existing optimizations for Multi-Rail networks, cost efficiency where the cost of each component is much cheaper, traffic isolation where each plane operates independently, latency reduction compared to a three layer fat tree, and robustness where multi-port NICs provide multiple uplinks so single port failures don't disrupt the entire GPU. Another optimization that the authors made was InfiniBand GPU Direct Async. Traditional IB operations required a CPU proxy thread where the GPU had to notify this proxy thread to populate the work request and signal the NIC. IBGDA allows the GPU to directly fill the work request and write to the RDMA MMIO address directly.

Limitations

The authors highlight some key limitations, most stemming from the limitations of the available hardware. Due to regulatory constraints, the H800s had reduced NVLink bandwidth restricting

efficient tensor parallelism and forcing MoE all-to-all communication onto slower InfiniBand links. Although FP8 mixed-precision training improves efficiency, current hardware imposes limited accumulation precision and high data-movement overhead for fine grained quantization, making it much more challenging to maintain numerical stability. Communication within the system also remains inefficient, as many tasks such as NVLink forwarding, data layout handling, type casting, and reduce operations must be executed on GPU SMs, consuming compute resources that should be used for model execution. While DeepSeek deploys a Multi-Plane Fat-Tree network, present NICs do not support true multi-plane packet spraying and require costly intra-node forwarding and adding extra latency. Reliability challenges also persist: intermittent NIC or NVLink failures, GPU crashes, and silent data corruption can disrupt long training runs, with limited built-in mechanisms for automatic detection or hardware-level recovery. Finally, despite KV compression, the transformer architecture remains memory-bound during autoregressive decoding, limiting long-context performance.

Future Research Directions

Across Sections 3 to 5, the authors propose improvements such as better support for low-precision computation, enhanced scale-up/scale-out interconnect convergence, and more efficient multi-plane networking topologies. In Section 6, the paper expands these ideas and offers concrete recommendations, beginning with advanced error-detection and correction capabilities to counter silent data corruption, including checksum-based validation and hardware-accelerated redundancy mechanisms. The authors also highlight future directions for alleviating CPU and interconnect bottlenecks by suggesting the adoption of direct CPU-GPU interconnects using NVLink.

Summary of Class Discussion

Q: How do you verify the correctness of those predicted tokens?

A: Not explicitly mentioned, but you can compare it to the baseline that doesn't use Multi-Token Prediction, if the tokens match then you can consider it as a success.

Q: Wondering whether or not it's adopted beyond deepseek and if yes or no what barriers are there?

A: Yes, they mention that they are not deployed in production. They don't employ them because their transformations might not work well in production. They want better transformations

Q: How worthwhile is it to invest in this hardware? Is it worth it to make hardware for specific models?

A: They don't have their own specialized hardware. They are orchestrating their hardware to get more efficient. They mention they offload a lot of network communications to the NIC as they want to fully utilize the gpu. At the end of the 2010s, people concluded that we have too much bandwidth and we should go back to oversubscription. Making changes at the network level isn't

a bad idea. At a micro level people are trying to build specific new hardware. Everyone is trying to specialize hardware that will be good at running certain models (transformers).

Q: For this hardware they are highly specialized to do matrix multiplication, even though it's so specialized for transformers, doesn't everything use matrix multiplication?

A: That's what you are assuming, that there will be a lot of matrix multiplication. In the future there could be something else. When change happens it will be big and rapid and we will wonder why we didn't have that before. Startups will die instantly when change happens, but big players are broad and diversified and will be able to move on.